

Viet-Sang Nguyen

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Experience

Secure-IC

Cryptography Engineer

Rennes, France

11/2025–present

- Analyzed potential attacks against cryptographic implementations and recommended countermeasures (masking, shuffling, redundancy).
- Assessed and supported cryptographic implementations to meet certification requirements (FIPS, SESIP, CC).

University of Saint-Étienne

Doctoral Researcher

Saint-Étienne, France

01/2023–10/2025

Thesis: Physical Attacks against Symmetric Cryptography Standards

- Initiated, led, and completed research projects resulting in 5 publications on side-channel and fault attacks.
- Developed and experimentally validated cryptographic attacks; designed and implemented effective countermeasures.
- Excellent grade in Post-Quantum Cryptography Engineering course (awarded 5 bug-bounty points for identifying errors in lectures).

CryptoExperts

Cryptography Engineer

Paris, France

09/2021–12/2022

- Designed and developed a generic framework for building circuits and generating white-box implementations of cryptographic algorithms.
- Designed and developed a framework for software obfuscation with a simple stateless secure element.

CryptoExperts

Cryptography Intern

Paris, France

03/2021–08/2021

- Designed and developed a cryptocurrency wallet with consideration of security protection.
- Studied white-box cryptography and physical attacks on ECDSA.

Research Institute XLIM

Cryptography Intern

Limoges, France

06/2020–08/2020

- Studied and developed an encrypted matching scheme.
- Improved speed by pre-discarding costly operations using Cuckoo filter.

Teaching

Embedded Programming (tutorials)

27h×3, for undergraduate students

IUT Saint-Etienne, France

2023–2024

C++ Programming (tutorials)

33h, for undergraduate students

IUT Saint-Etienne, France

2024

Publications

- **Nguyen**, Grosso and Cayrel, “*Practical Persistent Fault Attacks on AES with Instruction Skip*”, IACR CiC, 2025
- **Nguyen**, Grosso and Cayrel, “*Practical Second-Order CPA Attack on Ascon with Proper Selection Function*”, CASCADE, 2025
- **Nguyen**, Grosso and Cayrel, “*Correlation Power Analysis on Ascon with Multi-bit Selection Function*”, SECRIPT, 2025
- **Nguyen**, Grosso and Cayrel, “*SIFA on Nonce-based Authenticated Encryption: When Does It Fail? Application to Ascon*”, FDTC, 2025
- Mercadier, **Nguyen**, Rivain and Udovenko, “*Versatile Software Obfuscation from a Lightweight Secure Element*”, IACR TCHES, 2024
- **Nguyen**, Grosso and Cayrel, “*Provable Resistance of Threshold Implementations to S(I)FA*”, in submission

About Me

My main interest is cryptography engineering. I'm especially motivated by the cycle: analyzing how implementations can be attacked and using those insights to design and implement protections.

Education

PhD in Cryptography

2023-2025

University of Saint-Étienne

France

Supervisors: Vincent Grosso, Pierre-Louis Cayrel

Master in InfoSec & Crypto

2019-2021

University of Limoges

France

Magna cum laude

Bachelor of Engineering (CS)

2015-2019

HCM University of Technology

Vietnam

Magna cum laude, honors program

Scientific Reviewing

External reviewer for EUROCRYPT (2024,2025), TCHES (2026)

Honors/Awards

- 2025 1st Prize in NSUCRYPTO Olympiad
- 2024 1st Prize in NSUCRYPTO Olympiad
- 2019 Scholarship for 2 years of Master
- 2018 Award for Outstanding Students

Selected Projects

- 2025 Threshold Implementation of PRESENT cipher in ARM (3 weeks)
- 2024 Incremental 2nd-order CPA (3 weeks)
- 2021 Tutorials for DPA attacks (2 weeks)

Technical Skills

- ★ Skilled in both research and engineering.
- 🔒 Cryptography: symmetric, asymmetric, post-quantum
- Development: implementation, evaluation
- Attacks: side-channel, fault injection
- Countermeasure: design, analysis
- Standards and certification compliance
- Reverse engineering
- 🔗 Python, C/C++, Assembly, Java, GPGPU
- 🔗 Sagemath, ChipWhisperer, Valgrind, SCALib, Side-Channel Marvels

Languages

- English Full professional proficiency
- French Level B2
- Vietnamese Native

Reference

Dr. Vincent Grosso

CNRS Researcher at University of Saint-Étienne
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